



Neural Surrogate Model in an Extended Kalman Filter for Chlorine Concentration State Estimation in Water Distribution Systems

[Artelt, A.](#), Strotherm, J., Hermes, L., Hammer, B.

Bielefeld University, Germany

University of Cyprus, Cyprus



Water-Futures has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (Grant agreement No. 951424)

This research was supported by the Ministry of Culture and Science NRW (Germany) as part of the Lamarr Fellow Network. This publication reflects the views of the authors only.

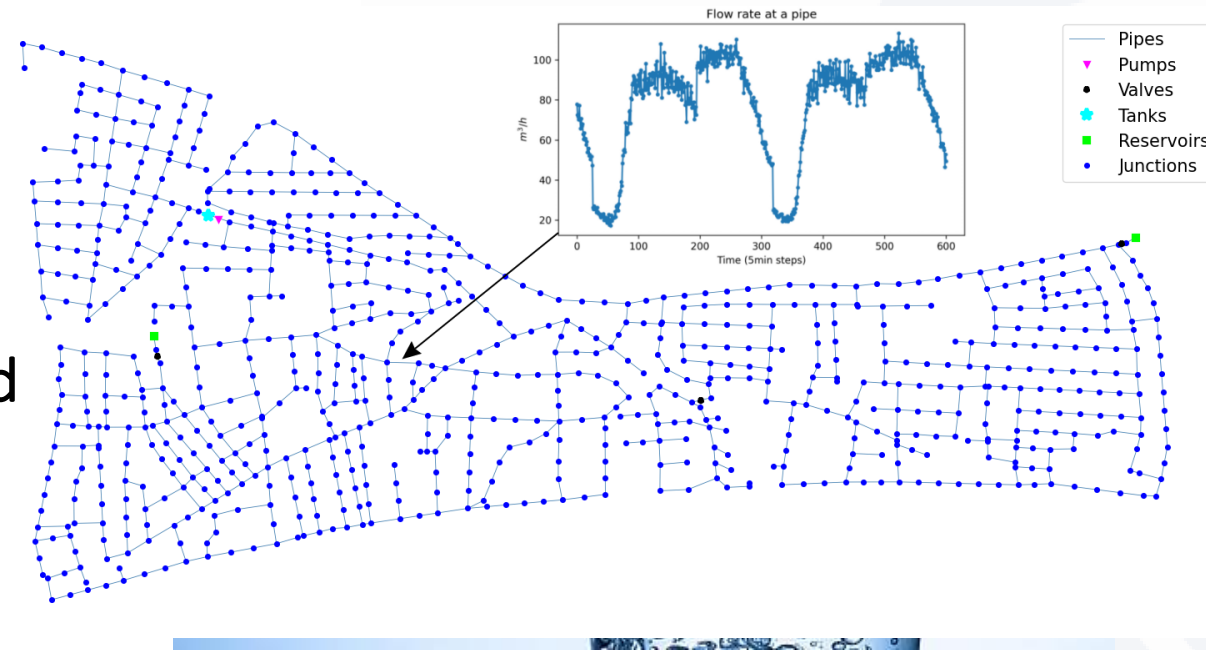
Water Distribution Networks (WDNs)

- Providing access to drinking water
 - Crucial for society => Critical Infrastructure
- Modeled as **graphs**
 - Complex **spatio-temporal signals**
 - Demands, Pressures, flows, chemical concentrations, etc.

=> Described by **differential algebraic and partial differential equations**

$$\frac{\partial f}{\partial q} = R (X2 + R (2 X3 + 3 R X4)) / |q|$$

$$\frac{\partial C_i}{\partial t} = -u_i \frac{\partial C_i}{\partial x} + r(C_i)$$



Tasks of Water Utilities

- Satisfy water demands at all times
- Ensure water quality – safe to drink
- Network planning
- Demand forecasting
- Leakage detection & localization
- Contamination detection & localization
- Control

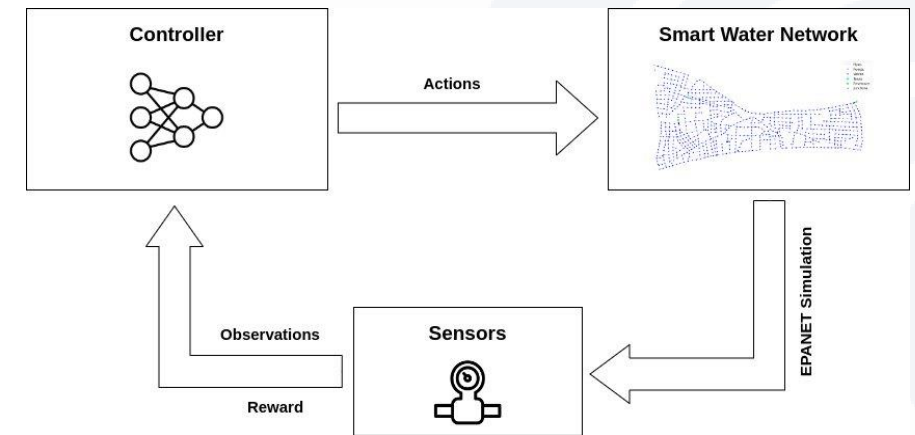
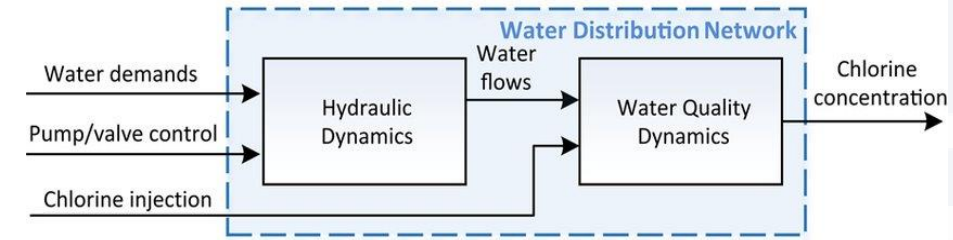


Water Chlorination Control

- Chlorine is a typical disinfectant
 - Mitigate contaminations
 - Too much chlorine is poisonous!
 - Disinfection By-Products
- Injection of Chlorine into the network
 - Maintain safe chlorine levels
 - Respond to contaminations & hydraulic changes
- Challenges:
 - Uncertainties
 - Sparse sensing
 - Unknown substances & reaction (parameters)
 - Complex dynamics
- How can AI help?

Potential of AI in Water Quality Control

- Surrogate models for complex dynamics
- **State estimation under uncertainty**
- Planning & Reinforcement learning for control
 - Hydraulic actuators
 - Chlorine booster stations
-



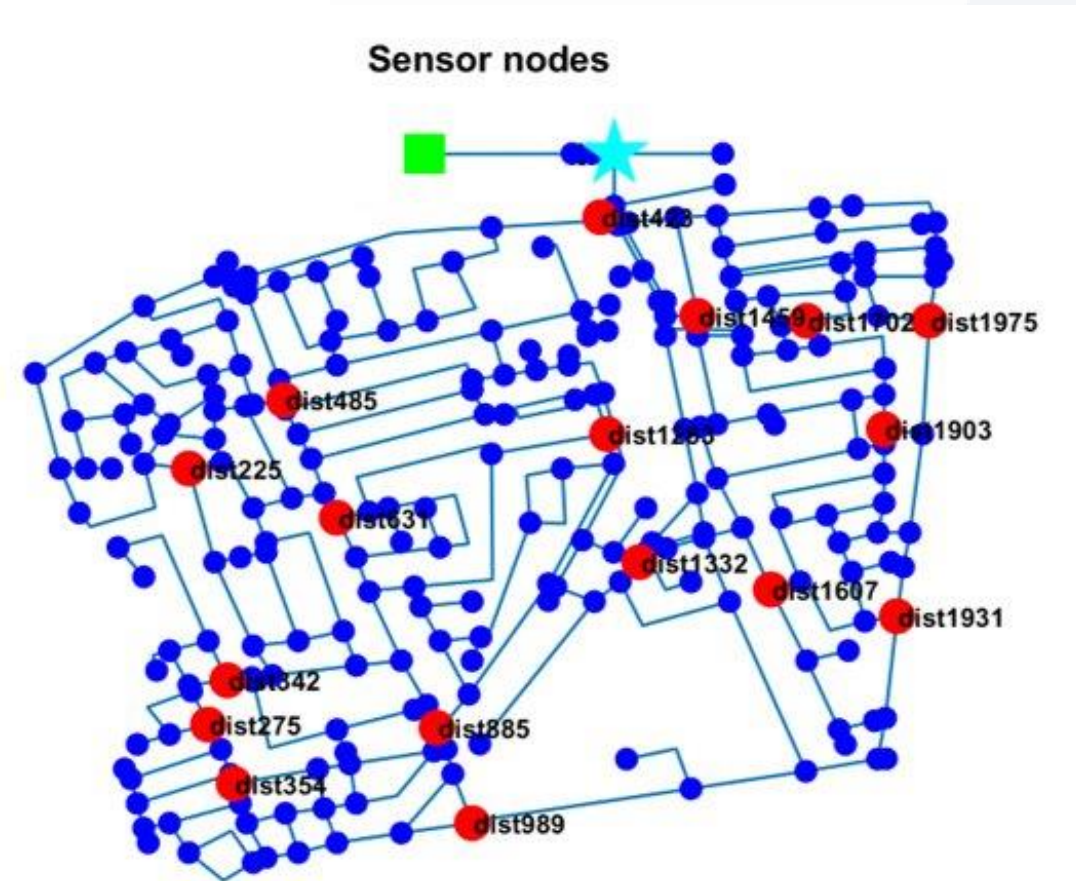
Water Quality State Estimation

- Sparse sensing
- Complex dynamics
- Uncertainties & Noise

=> Estimate states everywhere in the network

=> AI can help!

Combine AI with classic state estimation!



Our Methodology

$$\mathbf{y}_t = \mathbf{z}_t - h(\hat{\mathbf{x}}_t)$$

- Make use of existing methods
 - Extended Kalman

$$\mathbf{K}_t = \mathbf{P}_t \mathbf{J}_h(\hat{\mathbf{x}}_t)^\top (\mathbf{J}_h(\hat{\mathbf{x}}_t) \mathbf{P}_t \mathbf{J}_h(\hat{\mathbf{x}}_t)^\top + \mathbf{R}_t)^{-1}$$

$$\mathbf{P}_t = (\mathbf{I} - \mathbf{K}_t \mathbf{J}_h(\hat{\mathbf{x}}_t)) \mathbf{P}_t$$

$$\hat{\mathbf{x}}_t = \hat{\mathbf{x}}_t + \mathbf{K}_t \mathbf{y}_t$$

- Key challenge:

- Need Jacobian of the system (i.e., water quality dynamics)
- Simulators are not differentiable!

- Our approach:

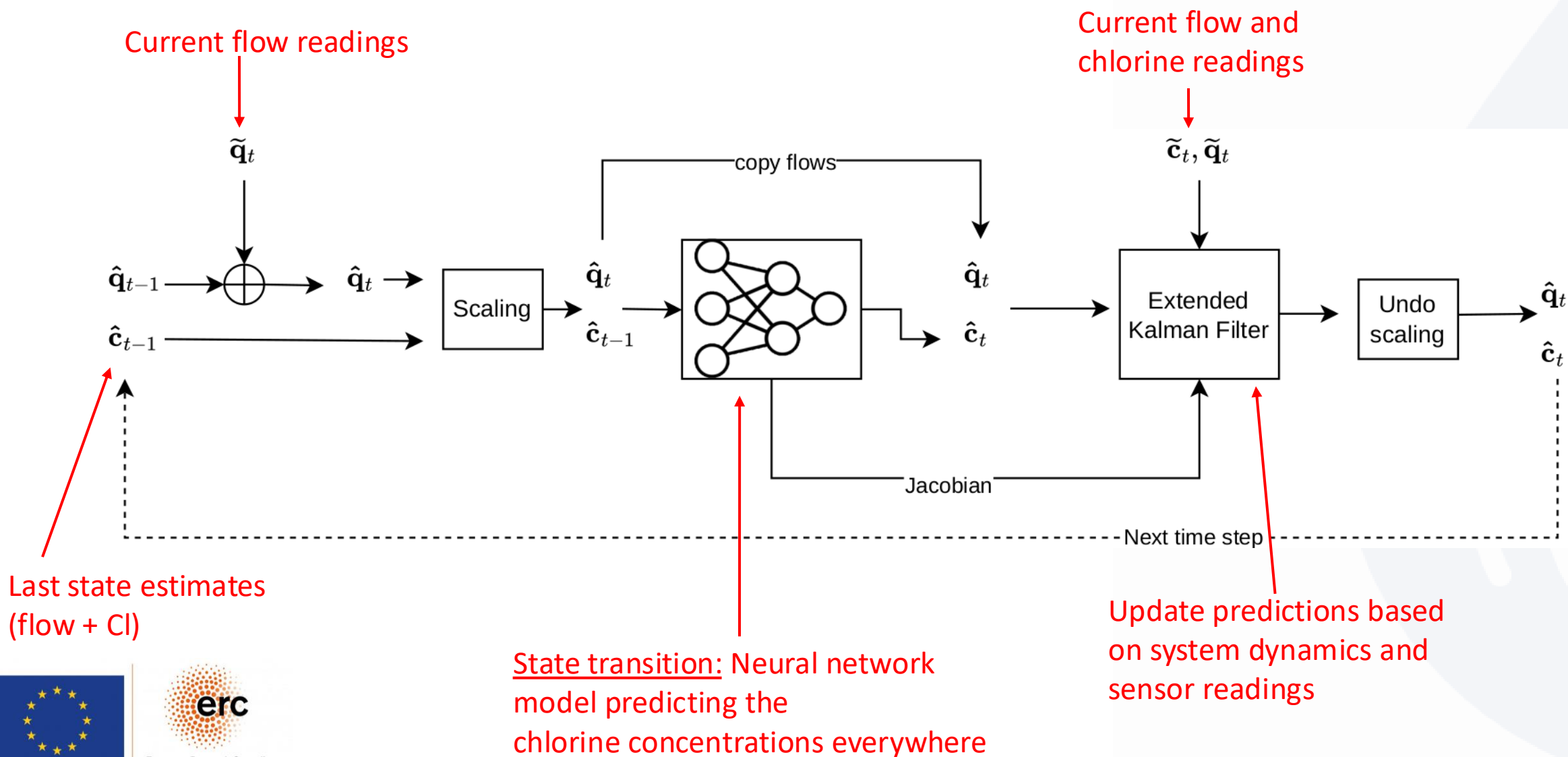
- Replace simulator by a differentiable neural network – i.e., surrogate model
 - Use its (time-varying) Jacobian

$\mathbf{z}_{t-1}$



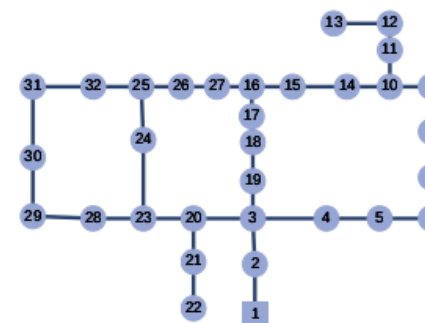
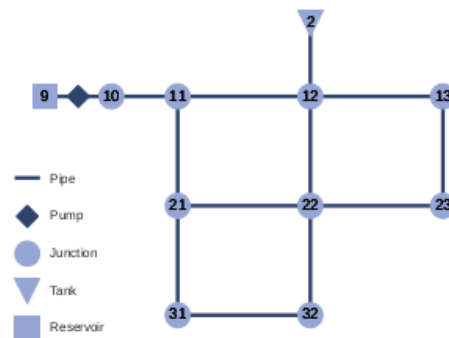
European Research Council
Established by the European Commission

Our Methodology

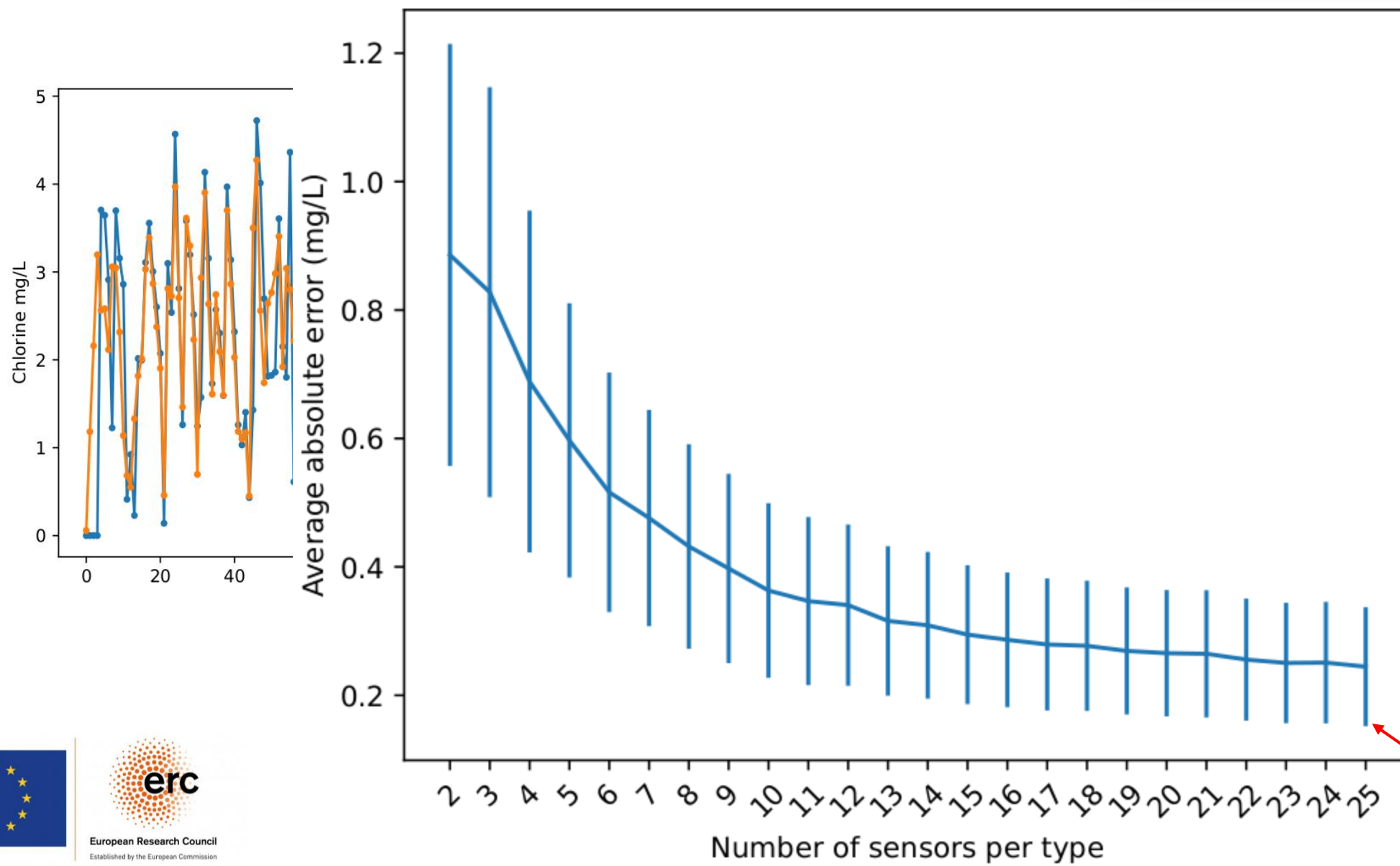


Experiments

- 3 benchmark networks from Literature
 - Net1, Hanoi, CY-DBP
- Randomized chlorine injection and demand
 - => Break any potential correlation!
 - => Key goal: Learn the physics
- Evaluate random sensor placements
 - => Placements have a huge effect on the performance!
- Compare to simulator's output



Results



Approx. 96%
"accurate" -- error of
4%

Summary & Outlook

- Water quality state estimation is important
- Many challenges (sparse sensing, uncertainties, complex dynamics)
 - => AI can help
- Our contribution:
 - *Physics-inspired Neural Network in an Extended Kalman Filter*
- Future work:
 - More advanced neural network architectures => Better scalability
 - Investigate & Reduce uncertainties
 - Integrate hydraulic surrogate
 -

